

that is from west to east.

12

B

$$\text{Container X: } p_X V_X = n_X RT_X \rightarrow p_X = \frac{n_X RT_X}{V_X} = \frac{n_X R 2T}{2V} = \frac{n_X RT}{V}$$

$$\text{Container Y: } p_Y V_Y = n_Y RT_Y \rightarrow p_Y = \frac{n_Y RT_Y}{V_Y} = \frac{n_Y RT}{V}$$

$$\text{Since } p_X = p_Y, n_X = n_Y, \frac{n_X}{n_Y} = 1.$$

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C